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The Anari and Tapirapuã Jurassic formations, western Brazil: paleomagnetism, geochemistry and geochronology

C.R. Montes-Lauar [a], I.G. Pacca [b], A.J. Melfi [b], E.M. Piccirillo [c], G. Bellieni [d], R. Petrini [c], R. Rizzieri [e]

[a] *Núcleo de Pesquisa em Geoquímica e Geofísica da Litosfera da Universidade de São Paulo, Caixa Postal 9638, 01065-970 São Paulo, Brazil*

[b] *Departamento de Geofísica, Instituto Astronômico e Geofísico da Universidade de São Paulo and Núcleo de Pesquisa em Geoquímica e Geofísica da Litosfera da USP, Caixa Postal 9638, 01065-970 São Paulo, Brazil*

[c] *Istituto di Mineralogia e Petrografia, Università di Trieste, Piazzale Europa 1, 34100 Trieste, Italy*

[d] *Dipartimento di Mineralogia e Petrografia, Università di Padova, Corso Garibaldi 37, 35100 Padova, Italy*

[e] *Centro di Studi Orogeno Alpi Orientali, c/o Dipartimento di Mineralogia e Petrografia, Università di Padova, Corso Garibaldi 37, 35100 Padova, Italy*

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Abstract

The Anari and Tapirapuã formations are very similar from the point of view of paleomagnetic, geochronological and geochemical results. They date from around 197 Ma and the flows are mainly tholeiitic basalts with a low TiO_2 and incompatible element content. The magnetic carriers in rocks from these two formations were highly oxidized titanomagnetites, maghemites and, probably, titanomaghemites. Paleomagnetic analysis has shown that the magnetizations are all normal in polarity and virtual geomagnetic poles obtained for both formations are indistinguishable at the 95% confidence level. The calculated mean for these poles is 250.3°E, 65.5°S ($N = 15$; $A_{95} = 3.6°$; $K_{SC} = 1578$). This pole is compatible with Jurassic poles which have been determined for South America or transposed from Africa.

1. Introduction

The tectonomagmatic reactivation of the South American Platform since the Permo–Triassic is related to the fragmentation of Gondwana and to the opening of the Atlantic Ocean. Magmatic episodes have occurred in continental and continental margin areas since the Permian and lasted until the Tertiary. The intensity and frequency reached a maximum during the Jurassic–Early Cretaceous. Two important magmatic periods, one during the Permo–Triassic and another in the Jurassic–Early Cretaceous are associated with the separation of Africa–South America and North America (opening of the North Atlantic) and the separation of Africa and South America (opening of the South Atlantic) ([1–4] and references therein). The magmatism related to these two periods is mainly tholeiitic, whereas during Late Cretaceous and Tertiary an alkaline character predominates.

A magmatism of tholeiitic character occurred in the sedimentary basins of Amazonas (3.4×10^5 km^3; [5]), Paraná (7.8×10^5 km^3; [6] and refer-

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ences therein) and Parnaíba, between 133–120 Ma [7–10] and 260–180 Ma. Dyke swarms are often found at the basin boundaries and sometimes in their interiors and they are distributed all over the country [8]. There are also local alkaline rock complexes of various lithologies. These mainly occur in the southern part of Brazil and reached a maximum frequency in the Late Cretaceous [8,11,12].

This article reports results on the study of magmatic bodies which occurred as a consequence of the Mesozoic reactivation of the South American Platform. Samples for this study came from the basalt formations of Anari in the State of Rondonia and Tapirapuã in the State of Mato Grosso. The motivation for the selection of these two formations was the scarce data on their paleomagnetism, geochronology and geochemistry, which made it impossible to conclude whether such basalts could belong to the flood tholeiites of the Early Cretaceous Serra Geral Formation (Paraná Basin), with which the basalts had previously been associated ([8], and references therein). Possible correlation with other magmatic bodies was also investigated. Further investigations on Ar–Ar dating, detailed petrography, geochemistry, mineral chemistry and Sr–Nd isotope are in progress.

2. Geological outline

The Anari and Tapirapuã basalts (and rare intrusives) occur in the Parecis–Alto Xingu sedimentary basin [13].

2.1. Anari Formation (Rondonia)

The Anari flows outcrop in the southeastern part of Rondonia State (Fig. 1), are tabular in

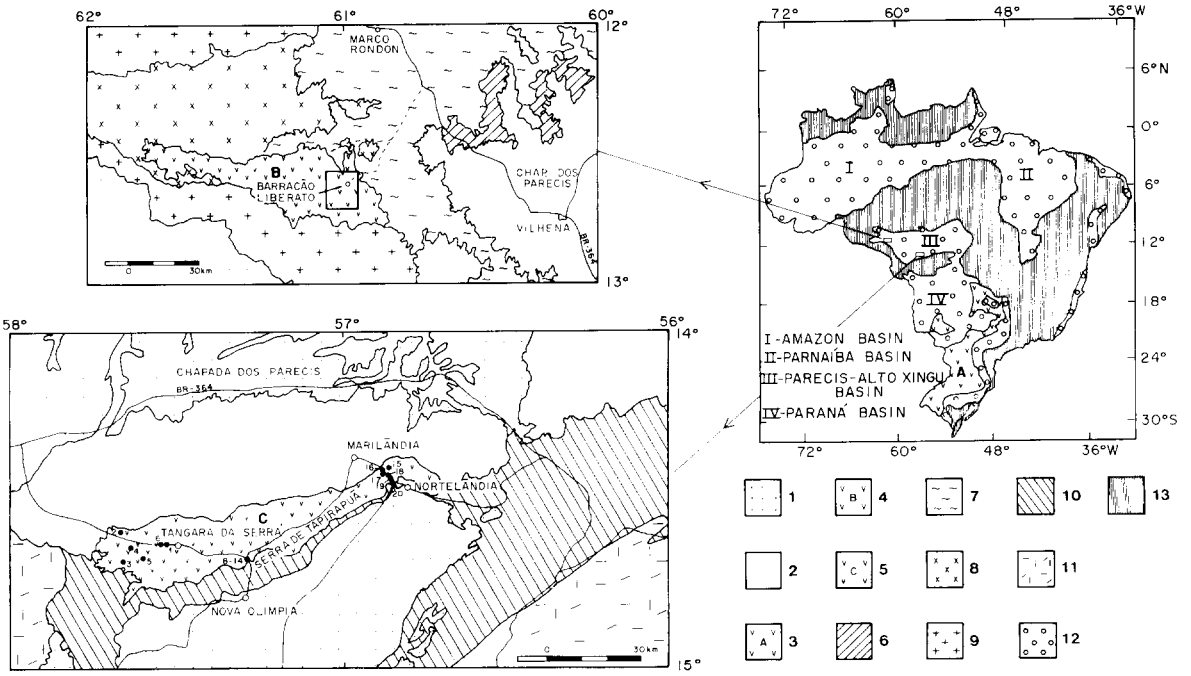


Fig. 1. Geologic map of the areas investigated. *B* = Anari Formation; *C* = Tapirapuã Formation (adapted from [14,15]). *1* = Cenozoic Cover; *2* = Parecis Group (Cretaceous); *3* = Serra Geral Formation (*A*) (Early Cretaceous); *4* = Anari Formation (Early Jurassic); *5* = Tapirapuã Formation (Early Jurassic); *6* = Botucatu Formation (?) (Triassic); *7* = Fazenda Casa Branca Formation (Permo–Carboniferous); *8* = Pimenta Bueno Formation (early Paleozoic); *9* = Xingu Complex (?) (Middle–Late Proterozoic); *10* = Alto Paraguai Group (Late Proterozoic); *11* = other Precambrian rocks; *12* = Phanerozoic; *13* = Proterozoic, Archean.

shape and elongated E–W (ca. 100 km long and 10–35 km across) [14]. They show a maximum thickness of 80 m in the western part and overlie gneisses and schists of the Xingu Complex (Middle–Late Proterozoic), sediments of the Pimenta Bueno Formation (early Paleozoic) and sediments of the Fazenda Casa Branca Formation (Permo–Carboniferous). The Anari basalts are remnants of flows which were fed by Mesozoic dykes (Amazonian craton), which were later partially eroded [8].

2.2. Tapirapuã Formation (Mato Grosso)

The Tapirapuã flows occur in south central Mato Grosso State (Fig. 1) and dip gently north ($< 3°$). They have an elliptical distribution and are ca. 115 km long, trending NE–SW, and 10–20 km wide. The thickness ranges from 310 m (southern part) to 35 m (northern part) [16,17]. These basalts overlie the sediments of the Alto Paraguay Group (Late Proterozoic) and are overlain by the Cretaceous Parecis Group and Quaternary sediments [15]. Some samples may have a significant amount of chlorite and celadonite, related to alteration processes. The best outcrops are found at the Serra de Tapirapuã.

3. Experimental procedure and results

3.1. Field work

Whenever possible, samples for paleomagnetic, geochronological and geochemical work were collected at the same sites. At least three oriented blocks were collected from each site for paleomagnetic work, using magnetic and sun compasses.

For Anari, 37 samples were collected from 12 sites for paleomagnetic analysis, 14 samples for geochemistry and 8 samples for dating. Only a relatively small region at the eastern side of the basalt unit was sampled (Fig. 1) due to access difficulties. The outcrops were usually small and surrounded by a dense forest, therefore the identification of flows and structural parameters was difficult but the flows seemed to be horizontal. The total thickness sampled was about 80 m, estimated from altitude measurements. From the Tapirapuã Formation, 61 samples (20 sites) were collected for paleomagnetism, 24 for geochemistry and 13 for dating. Easy access conditions and good outcrops with horizontal flows enabled a representative sampling, as shown in Fig. 1.

3.2. Radiometric age

Published K–Ar ages [14] of the Anari basalts range from 155 ± 2 to 143 ± 6 Ma (mean 151 ± 7 Ma) and one determination of 182 ± 3 Ma; the youngest age refers to a plagioclase concentrate. A new Ar–Ar age for Anari yielded a plateau age of 196.6 ± 0.4 Ma [P. Renne; pers. commun.]. This age is very well defined and, although based on a single sample, it is confirmed by the paleomagnetic pole corresponding to an Early Jurassic age, as will be discussed later. The Ar–Ar age is older than the K–Ar ages [14] but, considering the K–Ar ages as minimum values as well as the paleomagnetic results, the age of 197 Ma will be taken as representative for the Anari basalts.

Available K–Ar ages for Tapirapuã basalts range from 129 ± 4 to 117 ± 13 Ma ([18]; Hama, 1976 referred by [7]). Such basalts were usually believed to be coeval with those of the Paraná basin (Serra Geral Formation) [8,15]. New K–Ar ages (whole rock) for Tapirapuã basalts (Table 1) were determined at the Centro de Pesquisas Geocronológicas-USP (São Paulo). Although a relatively large dispersion of data is observed, there is agreement within experimental error between K–Ar ages obtained for this work and previous determinations (Table 1). Ar–Ar preliminary data [P. Renne, pers. commun.] for the Tapirapuã basalt indicate an age quite similar to that obtained with the same method for an Anari basalt (i.e., 197 Ma). As will be shown later, an Early Jurassic age for the Tapirapuã basalts is also consistent with paleomagnetic and geochemical data.

3.3. Petrological outline

The Anari tholeiitic basalts are fine to middle–coarse grained and slightly porphyritic. They

Table 1
K–Ar ages for the Tapirapuã formation

No. Sample	Rock Type	%K	^{40}Ar rad. ($x10^{-6}$) (ccSTP/g)	^{40}Ar atm. (%)	AGE (Ma)	
TRG1-6925	andesi-basalt	0.3007	1.50	54.28	124.4±7.1	
TRG6-6926	tholeiite	0.3385	1.64	42.69	120.8±6.4	
TRG7-6927	tholeiite	0.3377	1.84	39.03	135.1±8.6	
TRG8-6928	tholeiite	0.3586	2.12	58.46	146.3±9.5	
TRG11-6929	tholeiite	0.3510	1.81	48.19	128.2±8.9	
127	basalt	0.412	2.14	41.60	128.9±3.9	[18]
129	basalt	0.567	2.67	27.70	117.2±3.5	[18]
3123	basalt	0.371	1.89	79.00	126.3±13.3	[7]

rad. = radiogenic; atm. = atmospheric. All determinations were performed with whole rock. The age determinations of [7,18] were recalculated according to [19].

are essentially made up of plagioclase (An = 78–66%) and augite (Wo = 41–36%) phenocrysts and microphenocrysts of plagioclase (An = 66–46%), augite (Wo = 32%), pigeonite (Wo = 9%) and scarce magnetite (ULV = 47%). The matrix is composed of plagioclase, augite, pigeonite, abundant magnetite and sometimes quartz–alkali feldspar intergrowths.

The Tapirapuã basalts are generally finer grained than those from Anari, and usually contain glass (8–30 vol.%). They show a slightly porphyritic texture with plagioclase (An = 85–63%) phenocrysts and microphenocrysts of plagioclase (An = 67–50%), augite (Wo = 34%), pigeonite (Wo = 11–8%) and magnetite (ULV = 45–85%). The matrix contains the same minerals, as well as glass and rare quartz–alkali feldspar intergrowths.

The basalts studied are essentially tholeiitic (Fig. 2) [Bellieni et al., in prep.]. Some Tapirapuã samples plot in the fields of andesi-basalts and transitional basalts. The tholeiitic nature of the Anari and Tapirapuã rocks is also shown by the AFM diagram (inset Fig. 2) which is consistent with the mineral assemblages (e.g., coexisting augite and pigeonite).

The Anari and Tapirapuã basalts have a low TiO_2 content (1.10–1.42wt%) and a low incompatible element content. Variation diagrams for major and trace elements as a function of Mg# (Mg/Mg + Fe^{2+}; Fe_2O_3/FeO = 0.15) values (Fig. 3) show that the Anari and Tapirapuã rocks have a very similar composition and, in general, are chemically different from those with low titanium content (LTi) from the northern Paraná Basin (NPB)—Early Cretaceous—and from the Mosquito Formation in the Parnaíba Basin (MPB)—Jurassic, 180 Ma—([23,24] and references therein). The chondrite-normalized abundance patterns of rare earth elements [27] for Anari and

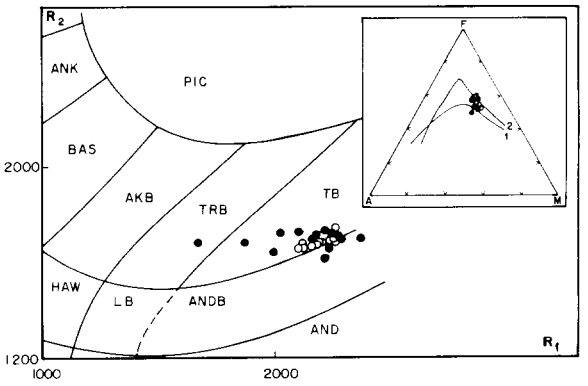


Fig. 2. R_1–R_2 diagram for Anari (○) and Tapirapuã (●) basalts [20,21]. PIC = picritic; ANK = ankaratrite; BAS = basanite; AKB = alkali-basalt; TRB = transitional basalt; TB = tholeiitic basalt; HAW = hawaiite; LAB = lati-basalt; ANDB = andesi-basalt. Inset: A(Na_2O + K_2O)–F(FeO_t)–M(MgO) diagram; *1* and *2* = alkaline and tholeiitic suites from Hawaii, respectively [22].

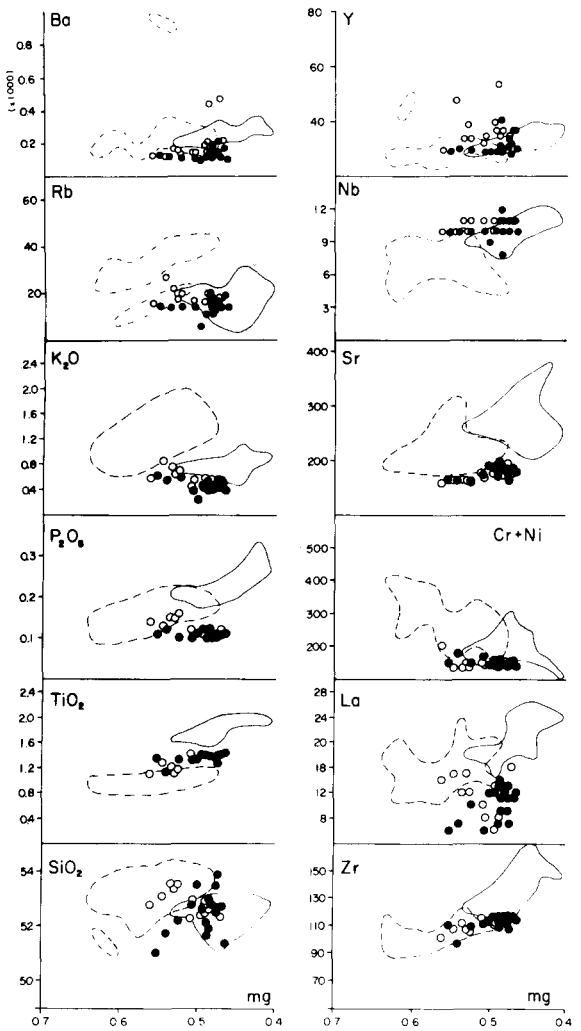


Fig. 3. Major (wt%) and trace (ppm) elements versus Mg# for Anari and Tapirapuã basalts. Symbols as in Fig. 2; solid outline = basalts from nothern Paraná Basin [23]; dotted outline = basalts from Mosquito Formation, Parnaíba Basin [24].

Tapirapuã basalts are similar and show a small fractionation (the mean La_n/Yb_n for Anari = 3.03 ± 0.15 and for Tapirapuã = 3.12 ± 0.04) and show small negative anomalies of Eu (mean Eu/Eu* for Anari = 0.89 ± 0.02 and for Tapirapuã = 0.88 ± 0.01). For the same MgO interval ($4.7 \leq MgO \leq 6.6\%$) the rocks from Anari and Tapirapuã show less fractionated patterns than those of LTi from NPB (mean $La_n/Yb_n = 4.45 \pm 0.26$; [25]) and those of LTi from MPB (mean $La_n/Yb_n = 4.50 \pm 0.42$; [26]). Sr and Nd isotope initial ratios (197 Ma) of Anari basalts range from 0.70531 to 0.70986 (mean = 0.70637 ± 0.00157) and from 0.51223 to 0.51245 (mean = 0.51233 ± 0.00010), respectively. These isotope variations are similar to those observed for the Tapirapuã basalts; that is $(^{87}Sr/^{86}Sr)_0 = 0.70566–0.70596$ (mean = 0.70579 ± 0.00012) and $(^{143}Nd/^{144}Nd)_0 = 0.51233–0.51242$ (mean = 0.51237 ± 0.00004). Most of these basalts plot close to the enriched quadrant of the mantle array, suggesting that they were not significantly affected by crustal contamination. Isotopically, the basalts studied are different from the LTi basalts of NPB [28] and from the Jurassic LTi basalts of MPB [24,26], which both have higher Sr isotope ratios.

3.4. Paleomagnetic work

3.4.1. Remanent magnetization (RM)

From every oriented block or hand sample, at least four, 2.5 cm diameter cores were cut and from these oriented cores at least three, 2.5 cm high cylindrical specimens were cut and used for paleomagnetic measurements. Spinner magnetometers were used for measurements: Molspin-MS1 and Schoenstedt DSM-2. Thermal and alternating field demagnetization procedures were, in general, similar for all samples. At least three specimens per sample were submitted to demagnetization treatment. The instruments used for AF demagnetization were the tumbling Molspin-MS2 and the stationary axis SI-4 Sapphire Instruments models. Thermal demagnetization was carried out with a Schoenstedt TSD 1 horizontal demagnetizer and changes in magnetic susceptibility were monitored with a Bison instrument model 3101A and a Sapphire S1.

One pilot specimen from every sample was submitted to detailed AF treatment in 5 mT steps, usually up to 80 mT. In some cases when the magnetization components were high in coercivity, fields up to 150 mT were applied. Another pilot specimen was submitted to detailed thermal demagnetization from 200°C to 650°C (in general) in steps of 50°C up to 500°C and above this temperature in 20°C or 30°C steps. The other specimens were submitted to at least 6 tempera-

Table 2
Paleomagnetic results for the Anari Formation

		Site mean direction					VGP's	
S	Nc/Nm	D(CaRM) (°)	I(CaRM) (°)	k	α_{95} (°)	R	LAT. (°S)	LONG. (°E)
4	3/3	10.5	-48.6	346.9	6.6	2.9942	70.4	270.7
5	3/2	25.6	-51.2	-	9.8	1.9985	59.7	252.5
6	4/2	27.5	-46.3	-	43.1	1.9720	60.2	243.5
7	3/3	17.8	-45.1	99.3	12.4	2.9798	68.2	251.7
9	3/2	25.5	-58.1	-	18.0	1.9949	55.4	262.7
12	3/2	10.6	-36.6	-	14.2	1.9968	77.3	247.8

	Direction of magnetization				Pole			
MEAN	D(°)	I(°)	k	α_{95}(°)	LAT(° S)	LONG(°E)	k	α_{95}(°)
6 sites	19.1	-47.9	84.1	7.3	65.5	254.9	79.1	7.6

s = site number; Nc/Nm = number of collected samples per site/number of samples used for mean determinations; D(CaRM), I(CaRM) = declination and inclination of the characteristic remanent magnetization, respectively; k, α_{95} = Fisher's parameters [30]; R = resultant vector.

Table 3
Paleomagnetic results for the Tapirapuã Formation

		Site mean direction					VGP's	
S	Nc/Nm	D(CaRM) (°)	I(CaRM) (°)	k	α_{95} (°)	R	LAT. (°S)	LONG. (°E)
1	3/3	25.5	-50.1	159.8	9.8	2.9875	61.5	251.7
7	3/3	21.9	-54.4	255.8	7.7	2.9922	61.7	262.4
8	3/2	20.7	-42.4	-	6.1	1.9994	68.2	242.7
9	3/3	20.2	-38.9	122.7	11.2	2.9837	69.5	236.5
10	3/2	21.9	-41.7	-	3.2	1.9998	67.4	240.4
11	3/1	22.7	-38.3	-	-	-	67.4	233.6
15	3/1	14.4	-40.4	-	-	-	73.9	247.8
18	3/2	26.6	-56.6	-	8.0	1.9990	57.1	262.1
19	3/2	30.7	-43.5	-	9.2	1.9987	59.2	238.9

	Direction of magnetization				Pole			
MEAN	D(°)	I(°)	k	α_{95}(°)	LAT(° S)	LONG(°E)	k	α_{95}(°)
9 sites	22.5	-45.2	114.5	4.8	65.5	247.2	128.0	4.6

s = site number; Nc/Nm = number of collected samples per site/number of samples used for mean determinations; D(CaRM), I(CaRM) = declination and inclination of the characteristic remanent magnetization, respectively; k, α_{95} = Fisher's parameters [30]; R = resultant vector.

ture or alternating field steps selected with pilot specimens. For most samples the optimum temperatures and/or alternating fields were between 400°C and 550°C and 30 mT and 40 mT, respectively. Viscous remanent magnetization (VRM) components were observed for most samples but they could generally be eliminated with 10–20 mT AF initial treatment.

The characteristic remanent magnetization (CaRM) per specimen was determined through the best grouping of directions obtained after several demagnetization steps, together with the corresponding Zijderveld diagram [29]. The CaRM per sample was calculated from the mean CaRM using all specimen values. Mean sample directions were then used to calculate site mean directions and the corresponding virtual geomagnetic poles (VGPs).

Paleomagnetic results for Anari and Tapirapuã are shown in Tables 2 and 3, respectively. All magnetizations are normal in polarity. Only sites with stable magnetization directions are repre-

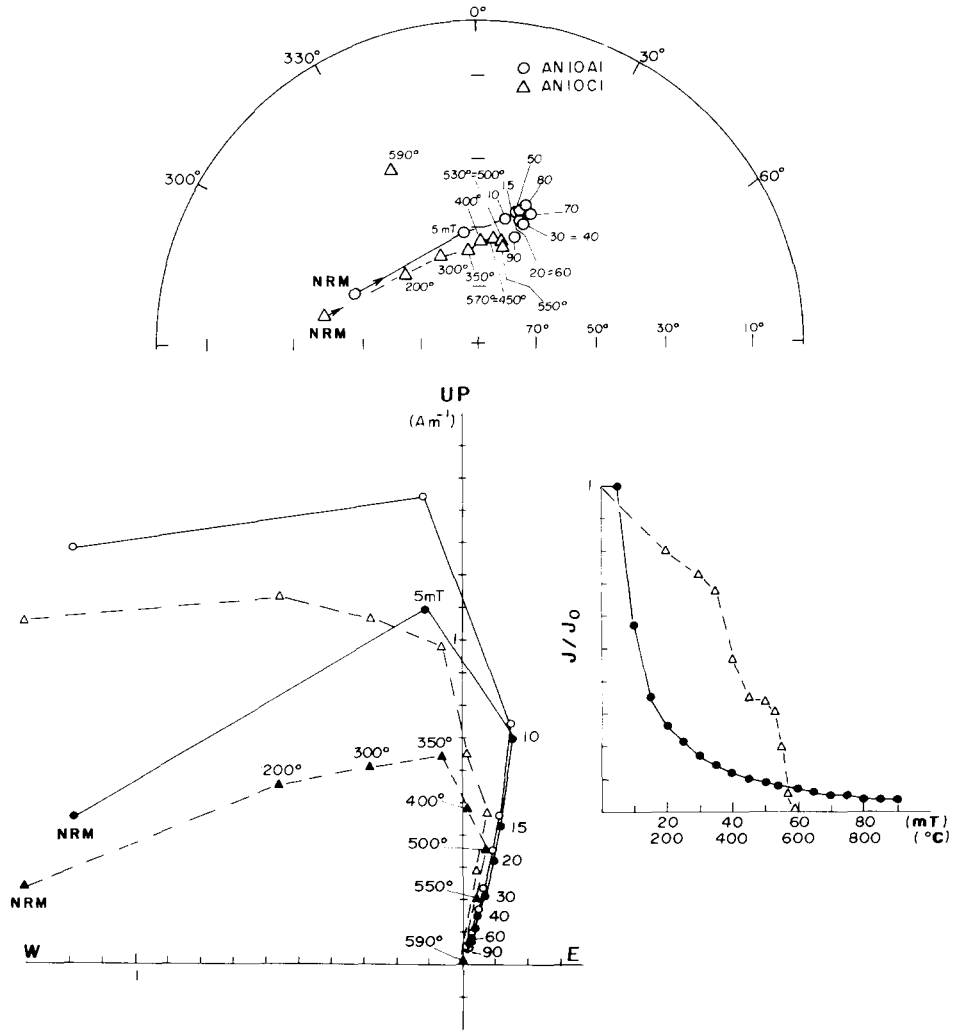


Fig. 4. Example of stable magnetization behavior for an Anari basalt. ○ and ● = AF demagnetization; ▲ and △ = thermal demagnetization. Upper figure: equal-angle stereographic projection, open symbols = normal polarity. Left: Zijderveld diagram, open symbols = vertical projections; solid symbols = horizontal projections. Right: normalized intensity curves.

sented. For several sites the magnetization direction was obtained with only two samples and, although in these cases the statistical quality of the mean cannot be estimated from the calculated α_{95} values, they have been included because, in general, their directions are coherent with the whole data set.

Part of the 37 Anari basalts samples (23) had to be discarded because it was impossible to find a stable direction or coherence among the site directions. Stable magnetization behavior for the Anari samples is shown in Fig. 4.

From the 61 Tapirapuã basalt samples only 19 (31%) yielded consistent results. One interesting feature of these rocks was that some sites presented very high NRM intensities, with an average of 99.9 Am^{-1} for eight sites for which no consistent results could be obtained. These NRM values are much higher than those obtained for the nine sites which defined the CaRM (6.4 Am^{-1}). It is probable that the anomalously high NRM intensities may have been caused by the superposition of an isothermal remanent magnetization. Stable magnetization behavior for the Tapirapuã rocks can be seen in Fig. 5.

3.4.2. Thermomagnetic analysis

At least one sample from each site was analysed for characterization of its magnetic constituents. A thermomagnetic balance was used for

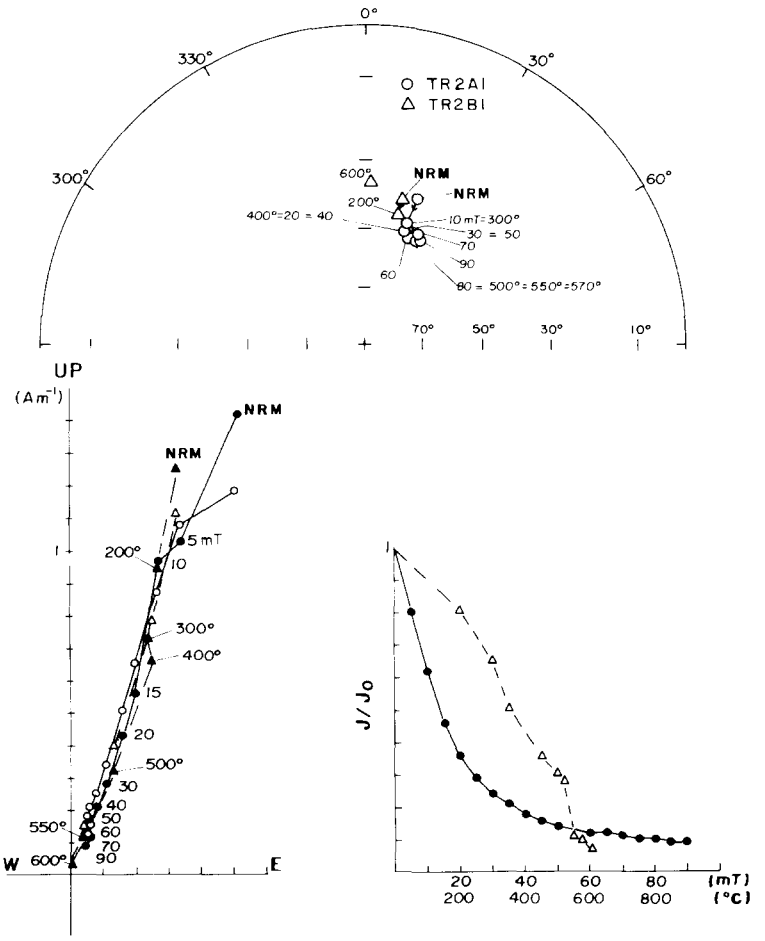


Fig. 5. Example of stable magnetization behavior for a Tapirapuã basalt. For explanation see Fig. 4.

the determination of Curie temperatures and the identification of magnetic minerals, complemented by optical microscopy.

Thermomagnetic analysis for the Anari basalt samples enabled the identification of two types of curves (Fig. 6A) where type a presents only one magnetic phase with high Curie temperature (CT, 561–602°C), indicating the presence of highly oxidized titanomagnetites [31]. The variation in CT observed for the samples indicates that the titanomagnetites have undergone variable degrees of oxidation. The type b curves (Fig. 6A) show two magnetic phases. The first one, with temperatures between 350°C and 358°C, characterizes the presence of maghemite, while the second (574–587°C) is an indication of titanomagnetites with low Ti content. It is interesting to analyze the blocking temperatures (BT) spectra in Fig. 6A, which were obtained from thermal demagnetization data for curves a and b. It may be seen that, although the magnetic mineral which has been identified by the analysis of thermomagnetic curves and $J/J_0 \times T$ curves is the same, BT spectra can be classified into two types: one that is thermally discrete (inset Fig. 6A below) and another that is thermally spread (Fig. 6A above; [32]), indicating that the samples contain magnetic grains with a diversity of sizes, anisotropies, etc., since the BT depends on these variables.

For the Tapirapuã basalts, four types of thermomagnetic curves were recorded (Fig. 6B). The type a curves are similar to those of type a of the Anari basalts, with highly oxidized titanomagnetites as the only magnetic phase (CT 490–605°C). The type b, c, and d curves are characteristic of titanomagnetites that have undergone low temperature oxidation ([33–35] and references therein). Titanomagnetites that have undergone a low temperature oxidation can be transformed into titanomaghemites, when thermomagnetic curves will be irreversible. Although curves of type b, c and d are typical of titanomaghemites they could also represent titanomagnetites with high Ti content. Samples corresponding to type b curves (CT 382–418°C; Fig. 6B) have, in general, not yielded favorable paleomagnetic results. A probable explanation is that the magnetic carriers (probably titanomaghemites) had low CTs and blocking temperatures (Fig. 6B, b) and were not able to preserve the primary remanent magnetization which, as indicated by data, corresponds to samples of type a, c and d curves. Thermomagnetic curves of type c and d, show a low CT (365–393°C) magnetic phase but it is not unique. The presence of titanomagnetites with low Ti content (CT of about 500–590°C) is probably an explanation for the magnetic stability of these samples.

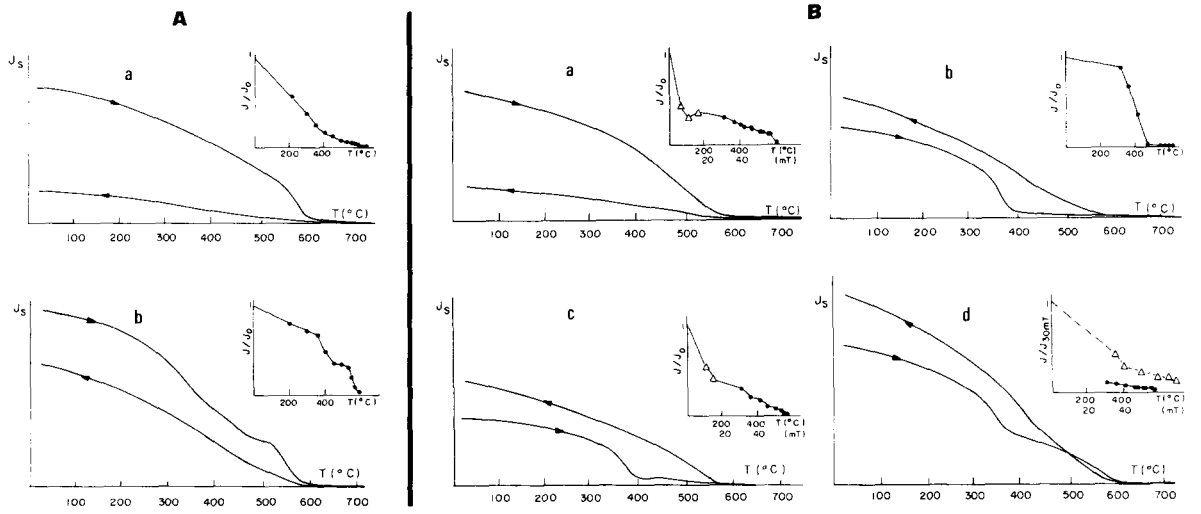


Fig. 6. Thermomagnetic curves for Anari (A) and Tapirapuã (B) basalts. Insets: normalized intensity curves.

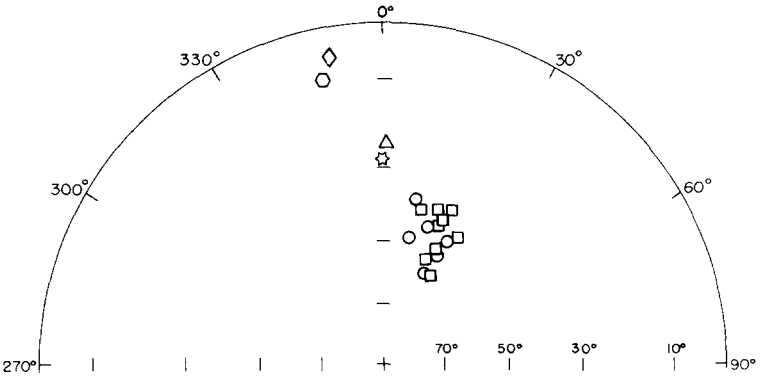


Fig. 7. Site mean directions for Anari (○) and Tapirapuã (□) basalts (all normal polarity). ○ = geomagnetic field for Tapirapuã basalts; ◇ = geomagnetic field for Anari basalts; ☆ = the dipolar field for Tapirapuã basalts; △ = dipolar field for Anari basalts (equal-angle projections).

3.4.3. Characteristic remanent magnetizations (CaRM), virtual geomagnetic poles (VGPs) and statistical tests

The paleomagnetic results for Anari and Tapirapuã presented in Tables 2 and 3 include individual paleomagnetic poles for the two formations. The site mean directions of CaRM for both formations are presented in Fig. 7 and Fig. 8 shows VGPs calculated for each sampling site.

As may be seen from Figs. 7 and 8, the positions of CaRMs and VGPs for the Anari and Tapirapuã samples on the stereonets are quite similar. This observation led to the suggestion that the two magmatic events which gave rise to these basalts could be associated, although this is not in agreement with previous K–Ar age determinations for Anari (mean age 151 Ma; [14]) and Tapirapuã (mean age 126 Ma; [18]; Hama, 1976 referred to in [7] and results from this work, Table 1). However, the paleomagnetic results have been confirmed by Ar–Ar dating and similarities between the two formations are also shown in chemical and isotopic analyses [Bellieni et al.; in prep.].

The statistical procedure of [36] was applied to estimate whether the VGPs obtained for Anari and Tapirapuã represented independent directions at the 95% confidence level (Table 4). The results indicate that the two sets of VGPs cannot be considered as independent at such a confidence level. Although the relatively small number of data available for Anari and Tapirapuã sets some limitations on the reliability and interpretation of data, it was concluded that the statistical tests confirm the evidence from other data that there is a relationship between the two forma-

Table 4
Comparisons of VGPs of Anari (1) and Tapirapuã (2) formations

N_1	N_2	R_1	R_2	R	k_1	k_2	A	B	(1-p)x100
6	9	5.93681	8.93749	14.86870	79.12826	127.96880	0.04018	0.25916	40.1

N = number of VGPs; R_1 and R_2 = vector resultant of 1 and 2, respectively; R = vector sum, $R_1 + R_2$; k_1 and k_2 = Fisher's parameters [30].
$A = r[(R_1 + R_2)^2 - R^2]/2[(N_1 - R_1) + r(N_2 - R_2)](R_1 + rR_2)$; $p = 0.05$; $r = k_2/k_1$.
$B = (1/p)^{1/N-2} - 1$; $p = 0.05$; $N = N_1 + N_2$.
The last column presents the probability of distinctness between the VGP groups.

Table 5
Mean direction of magnetization for Anari and Tapirapuã and paleomagnetic pole

Mean direction of magnetization					Paleomagnetic pole				
No. of Sites	Dm (°)	Im (°)	k	α_{95} (°)	No. of Sites	Lat. (°S)	Long. (°E)	K_{SC}	A_{95} (°)
15	21.2	-46.3	102.8	3.8	15	65.5	250.3	1578.1	3.6

Dm, Im = mean declination and inclination, respectively; k, α_{95} = Fisher's parameters [30]; K_{SC}, A_{95} = McFadden's parameters [37].

Table 6
Mesozoic poles for South America and Africa

No.	Location	Age (Ma)	N	Lat. (°S)	Long. (°E)	α_{95} (°)	Ref.
01	Almafuerte lava flows - Argentina	120-130	15	72	25	6	[38]
02	Ponta Grossa dykes - Brazil	> 120	115	82	30	2	[39]
03	Serra Geral flows (SG2) - Brazil	132	28	82	38	8	[4]
04	Ceará Mirim dykes (I,V) + Sardinha intrusions (MC1) - Brazil	125-145 120-130	29	84	90	2	[40]
05	Sierra de Los Condores Group - Argentina	121	08	84	91	5	[41]
06	Serra Geral flows (SG1) - Brazil	135	258	85	108	1	[4]
07	Arqueros igneous rocks - Chile	120-130	06	81	165	11	[42]
08	Chon-Aike - Argentina	166	26	84	243	10	[43]
09	Ceará Mirim dykes (III,IV) + Mosquito flows (MC2) - Brazil	160-175 170-200	39	83	215	4	[40]
10	Draa Valey, Saara - Maroc	180-186	16	75	236	4	[44]
11	Liberia dykes - Liberia	173-192	25	70	231	5	[45]
12	Stormberg flows, Sani Pass and Maseru - Lesotho	154-190	(54)	54	214	5	[46]
13	Mateke and Marangudzi Compl.- Rodesia	190±10	10	62	207	7	[47]
14	Bolivar dykes - Venezuela	198	05	67	246	4	[48]
15	Anari and Tapirapuã flows - Brazil	197	15	66	250	4	this paper
16	Meseta and Atlas - Maroc	180-205	34	70	257	6	[49]
17	Las Cabras igneous rocks - Argentina	204	05	74	266	13	[50]
18	Guacamayas Group - Venezuela	195	05	70	301	12	[48]
19	Ceará Mirim dykes (II,RN2) - Brazil	Early Jurassic	27	64	305	4	[40]
20	Suriname dykes - Suriname	227	10	82	320	10	[51]
21	Ischigualasto-Ischichuca - Argentina	224	09	79	239	15	[52]
22	Puesto Viejo - Argentina	232	06	76	236	18	[52]
23	Guyana Dolerites	262-675	05	63	222	12	[53]

Data from [4,38–53] and this paper.
No. = corresponding number in Fig. 9; N = number of the sites, () = number of the samples; α_{95} = Fisher's parameters [30].

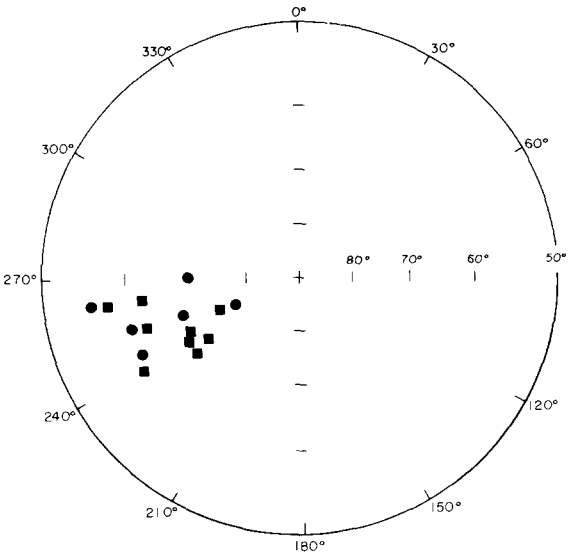


Fig. 8. Southern geomagnetic virtual poles for Anari (●) and Tapirapuã (■) formations (equal-angle projections).

tions. Therefore, in view of all the evidence, a single paleomagnetic pole has been calculated for Anari and Tapirapuã (Table 5).

4. General discussion of paleomagnetic results

The framework of South American Mesozoic paleomagnetic poles (Table 6 and Fig. 9) needs a better definition which can only be obtained by increasing the amount of good quality data. Many poles have been calculated with a small number of sites and even when the number of sites was reasonable there may be poles for which the effects of secondary magnetizations and secular variation have not been completely eliminated. Another critical factor is dating, since most determinations are K–Ar ages, which have serious limitations.

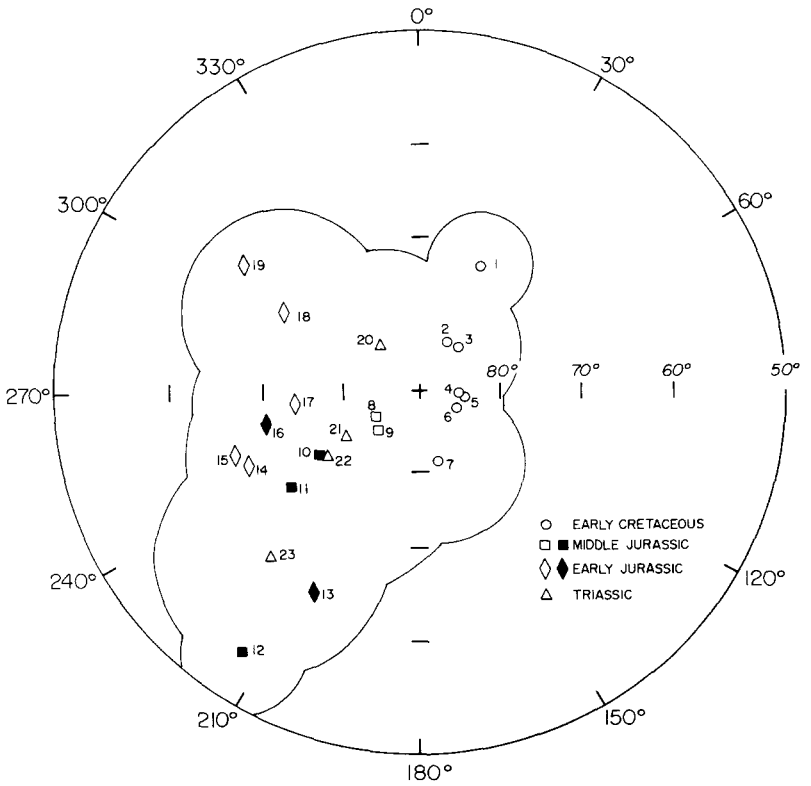


Fig. 9. Mesozoic poles for South America (open symbols) and Africa (solid symbols). For identification of poles see Table 6. Adapted from [40] (equal-angle projections).

The previous Early Jurassic paleomagnetic poles for South America (Table 6; Fig. 9) were determined using only 5 sites each and are rather dispersed. The pole determined in this work for the Anari and Tapirapuã formations (Anari + Tapirapuã; no. 15, Table 6; Fig. 9) agrees with the one obtained for the Bolivar dykes: 198 Ma (no. 14, Table 6; Fig. 9). The age which has been attributed to the Anari + Tapirapuã pole is 197 Ma. The pole calculated for the Anari and Tapirapuã formations also agrees with Jurassic African poles, which have been determined for the Meseta and Atlas [49], for the Draa Valley sills, the Sahara shield [44] and for Liberian dykes [45] (Table 6; Fig. 9, in a pre-drift reconstruction, [54]).

The Anari + Tapirapuã paleomagnetic results, supported by other evidence, therefore suggest that these two formations correspond to magmatic events which occurred long before the Early Cretaceous Serra Geral Formation flood basalts (no. 6, Table 6; Fig. 9).

5. Conclusions

The Anari and Tapirapuã formations are composed mainly of tholeiitic basalts. The minimum K–Ar age obtained for the Tapirapuã Formation was 126 Ma. The Ar–Ar age determined for Anari and Tapirapuã was 197 Ma. This value (197 Ma) was then adopted for both formations. The rocks of both formations are characterized by low contents of TiO_2 and incompatible elements. The geochemical and isotopic analyses show that the rocks from Anari and Tapirapuã are similar and, in general, they are different from the of low TiO_2 content rocks of the Northern Paraná Basin (Early Cretaceous) and the Mosquito Formation at Parnaíba Basin (Middle Jurassic). Sr and Nd isotopic initial ratios (197 Ma) suggest that rocks from the Anari and Tapirapuã formations have been slightly contaminated by crustal material.

Only 31% of the samples from Tapirapuã and 40% of the samples from Anari yielded coherent magnetization directions. This fact may be related not only to the magnetic mineralogy of rocks but also to probable isothermal remanent magnetization components which affected the rocks. The thermomagnetic analysis of the Anari basalts indicated that the magnetic carriers are mainly highly oxidized titanomagnetites and subordinate maghemites. For the Tapirapuã rocks, highly oxidized titanomagnetites and probably titanomaghemites are also observed. Sampling site characteristic remanent magnetizations for Anari and Tapirapuã yielded normal polarity. The paleomagnetic analysis and statistical treatment showed that the virtual geomagnetic poles found for Anari and Tapirapuã are indistinguishable at the 95% confidence level. The paleomagnetic pole calculated (250.3°E; 65.5°S; $N = 15$; $A_{95} = 3.6°$; $K_{SC} = 1578.1$) is compatible with another South American pole of Jurassic age and with African poles.

The association of paleomagnetic, geochronological, geochemical and isotopic results suggests that the Anari and Tapirapuã formations are associated in time and chemical composition. The results show that the Tapirapuã and Anari formations cannot be considered as extensions of the Serra Geral magmatism in the Paraná Basin (Early Cretaceous). The age that has been found for both formations (197 Ma) and the calculated paleomagnetic pole identify a magmatic event which preceded the great Early Cretaceous episode in the Paraná Basin (Serra Geral Formation) and may be related to the early phases of the South American Platform tectonic reactivation which started in the Permo–Triassic (South Atlantian event).

On the other hand, the age and location of Anari and Tapirapuã is compatible with the hypothesis that these formations originated from the same mantle thermal anomalies which, about 70 Ma later, gave rise to the Serra Geral magmatism. A possible relation between these magmatic events and hot spots such as Tristan da Cunha and Gough will be investigated further.

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